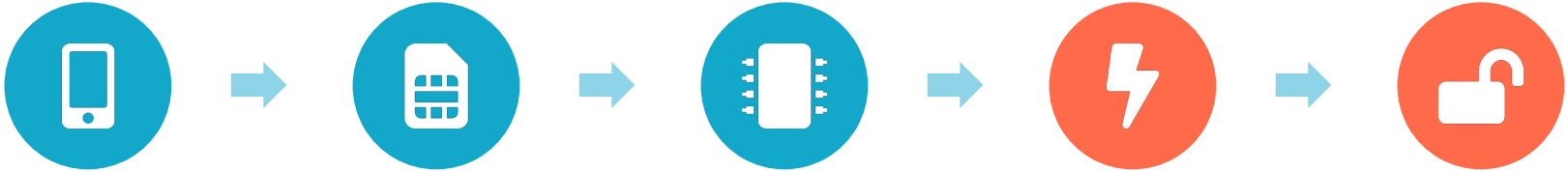
BUILD GUIDE

Pay-to-Unlock Pool Gate

An M-Pesa payment of exactly KSh 20 releases the gate lock. ESP32 + SIM900 + solenoid.



How the system works



Customer pays

Sends KSh 20 by M-Pesa to the SIM's number

SIM900

Receives the M-Pesa SMS over 2G

ESP32

Reads it with AT commands and checks it

Relay

Switches 12 V for a short pulse

Solenoid

Latch releases, then re-locks

KSh 20.00

exact amount, nothing else

1 code = 1 unlock

repeat SMS are ignored

3 seconds

then the lock re-secures

What you need to buy

ESP32 DevKit
30 or 38 pin, plus a USB cable for flashing

SIM900 board
GSM/GPRS shield or breakout, with antenna

SIM card
A registered M-Pesa line with the PIN turned off

12 V solenoid
Fail-secure latch, 0.5 to 1 A, IP-rated

Relay module
5 V, 1 channel, opto-isolated (or a MOSFET)

12 V power supply
3 A or more. Runs everything

Buck converter
12 V to 5 V, 3 A (LM2596 type)

Small parts
1N4007 diode, 1k and 2.2k resistors, 1000 µF cap, IP65 box

Power plan

12 V

Main supply. Drives the solenoid and feeds the buck converter

5 V

From the buck converter: ESP32, SIM900 and relay module

2 A

Peak current the SIM900 can pull when it connects to the network

Do

- Join all grounds together
- Put a 1000 µF capacitor at the SIM900 supply
- Attach the antenna before powering on

Don't

- Power the SIM900 from the ESP32 pins
- Put 12 V on any 5 V pin
- Plug USB and 5 V into VIN together unless the board has a diode

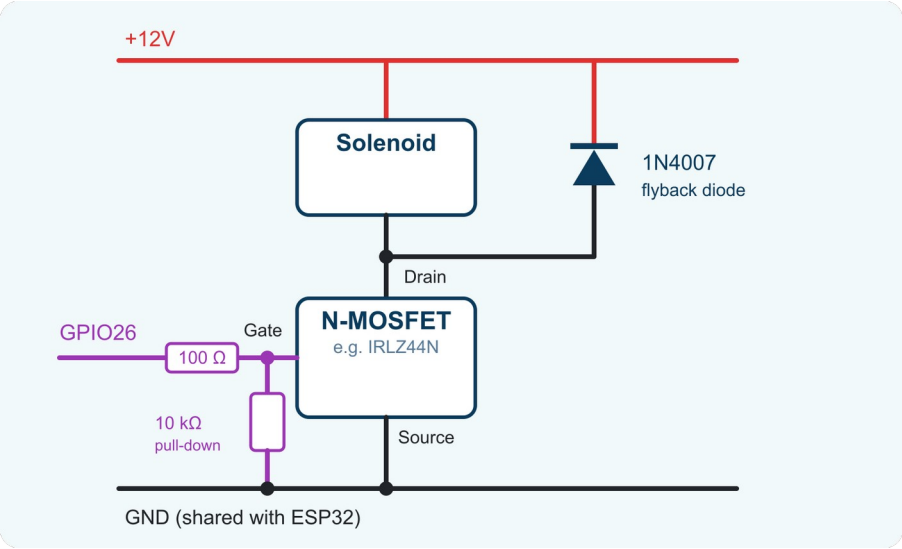
Wiring: SIM900 to ESP32

SIM900	ESP32	Note
TXD	GPIO16 (RX2)	Direct wire
RXD	GPIO17 (TX2)	Through a 1k / 2.2k divider
GND	GND	Must be common
5V in	Buck 5 V	Never from the ESP32
PWRKEY	Optional GPIO	Only if the board needs it

Serial settings: 9600 baud, 8N1. The firmware sets this up for you.

ESP32-WROVER? GPIO16/17 are taken by PSRAM. Change the pins in config.h.

Wiring: solenoid driver



MOSFET option

Relay module (default)

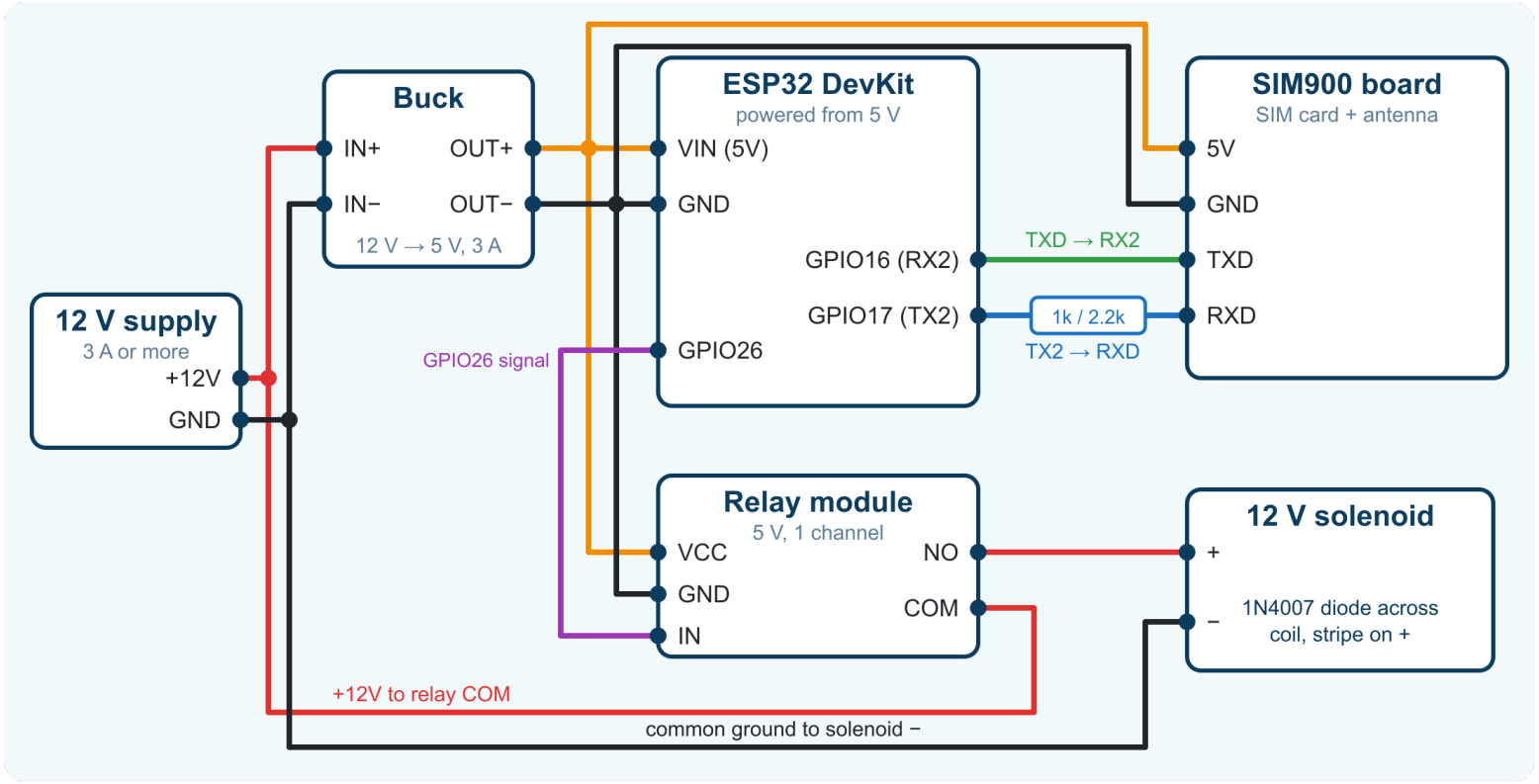
- Simplest, any 12 V solenoid
- Many modules trigger on LOW: set LOCK_ACTIVE_HIGH to 0

MOSFET option

- Silent, no contact wear
- Needs a logic-level part and the 10k pull-down

A flyback diode across the coil is required with either option.

Full wiring diagram



Prepare the SIM card

1. **Turn off the SIM PIN**
   Put the SIM in a phone and disable the PIN lock. The ESP32 does not send a PIN.

2. **Use a registered M-Pesa line**
   Customers send money to this number, so it must be an M-Pesa-registered SIM.

3. **Check 2G coverage**
   The SIM900 is 2G only. Test the signal at the pool before you build.

4. **Send yourself KSh 20**
   From another phone. The SMS must read: You have received Ksh20.00 from.

Flash the firmware over USB

1

Install Arduino IDE

Add the esp32 board package by Espressif in Boards Manager.

2

Open the sketch

firmware/pool_lock/pool_lock.ino. Keep all three files together.

3

Pick board and port

Tools: ESP32 Dev Module and the COM port of your ESP32.

4

Check config.h

Pins, unlock time and the amount (EXPECTED_CENTS = 2000).

5

Upload

Press Upload. Hold the BOOT button if it says Connecting.

6

Open Serial Monitor

Set 115200 baud and watch the start-up messages.

Test before you install

1. Power on and read the Serial Monitor
2. Type OPEN: the relay clicks
3. Type SIM: plus a sample M-Pesa SMS
4. Send a real KSh 20 payment
5. Send KSh 10 or 30: it must be rejected

```
[GSM] registered on network
[GSM] signal: +CSQ: 18,0
[SYS] ready, waiting for payments

[SMS] VALID payment SDE1ABC234
[LOCK] UNLOCK for 3000 ms
[LOCK] locked again

[SMS] rejected: wrong amount
[SMS] duplicate transaction
```

What counts as a valid payment

Accepted

- Sender ID is MPESA
- Starts with a 10-character code
- Says Confirmed and You have received
- Amount is exactly Ksh20.00
- Code has not been used before

Rejected

- Ksh19.99, Ksh20.01, Ksh200
- Money sent or paid out
- The same code a second time
- Any other sender
- Balance, reversal and promo texts

Known limit: a sender ID can be spoofed. For stronger proof, confirm payments through the Safaricom Daraja API (needs internet).

Safety and troubleshooting

12 V DC only near water

Keep electronics in an IP65 box, away from splashes.

Keep a manual override

A key or release so nobody is locked out if power fails.

No reply to AT

Swap TX and RX, check 9600 baud and the common GND.

Board resets on network

SIM900 power is too weak: 5 V 3 A plus the capacitor.

Not registered

Check antenna, 2G coverage and that the SIM PIN is off.

Relay works backwards

Set LOCK_ACTIVE_HIGH to 0 in config.h.

Questions for you

Which ESP32 board?

WROVER boards need different UART pins.

Solenoid voltage and current?

I assumed 12 V and up to 1 A.

Relay or MOSFET?

The code works with either.

How do customers pay?

I assumed send-money to the SIM's own number.

Unlock time and repeats?

Now 3 s, and a second payment restarts it.

Is 2G available on site?

Otherwise a 4G modem such as the A7670 is safer.